SENTINEL-GNSS

Proactive Multi-Horizon Prediction of GNSS Signal Degradation
for Autonomous Vehicle Navigation

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1. The Problem

Autonomous vehicles at 60 km/h travel **17 metres every second**. GNSS—the sole source of drift-free absolute position—fails silently in urban canyons: buildings reflect or block satellite signals, producing positioning errors of **10–200 m** with no external indication.

Every existing monitor is reactive. RTKLIB quality codes, RAIM, and recent ML classifiers report degradation *at the moment it happens*, giving the vehicle **zero** preparation time.

We ask: can degradation be predicted 5–30 seconds before it occurs?

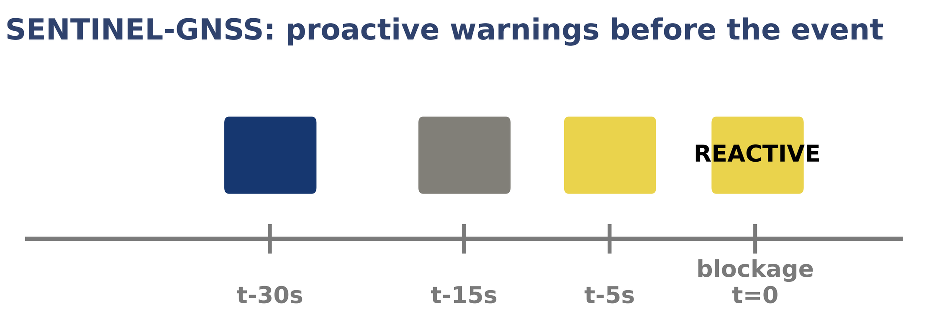

Figure 1: Reactive vs. proactive degradation warning. SENTINEL raises WARNING before the vehicle enters the NLOS zone.

2. What a Warning Buys

Horizon	Distance	Vehicle action
+5 s	83 m	Tighten IMU fusion, slow down
+15 s	250 m	Pre-engage dead-reckoning
+30 s	500 m	Re-route to avoid NLOS zone

3. Dataset

149,662 labelled epochs · 4 cities · 9+ receiver types

- Beihang campus (Scenarios A–E): controlled blockage scenarios
- UrbanNav Hong Kong: urban canyons, tunnels, open roads
- **Tokyo (held out):** Shinjuku + Odaiba — never seen during training

3-class labels: CLEAN / WARNING / DEGRADED

37 features/epoch in 7 groups: position, signal strength, satellite geometry, pseudorange, temporal, atmospheric, receiver status

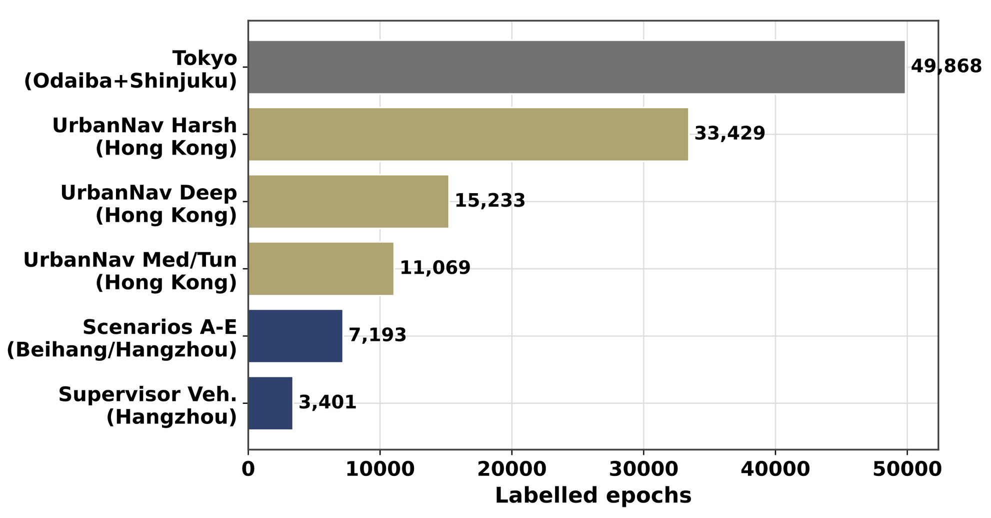

Figure 2: Dataset composition by city and receiver type.

4. SENTINEL-GNSS Architecture

Hybrid Transformer + BiLSTM (1.46 M parameters)

- **Transformer encoder** (2 layers, 4 heads, $d = 128$): captures global satellite-geometry trends across the 60-second window
- **BiLSTM** (2 layers, hidden 256): captures sequential approach dynamics (gradual C/N_0 decline)
- **Three output heads:** +5 s, +15 s, +30 s
- **Training:** Focal loss ($\gamma = 1.0$), class weights $[1, 2, 5]$, AdamW, cosine annealing

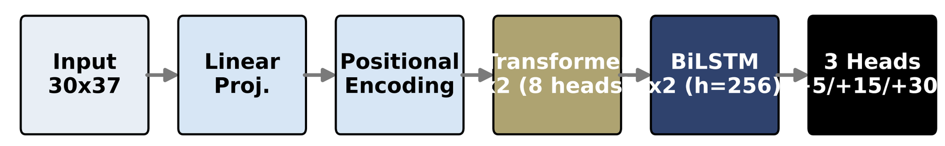

Figure 3: SENTINEL architecture: one forward pass produces all three prediction horizons.

5. Classification Results

Horizon	Macro-F1	MCC	DEG F1
+5 s	**0.821**	0.773	**0.718**
+15 s	0.741	0.691	0.583
+30 s	0.783	0.731	0.629

DEGRADED recall at +5 s: 0.847 — catches 85 % of impending degradations 5 seconds early.

10.5× faster than three separate tree models; 0.039 ms/sample.

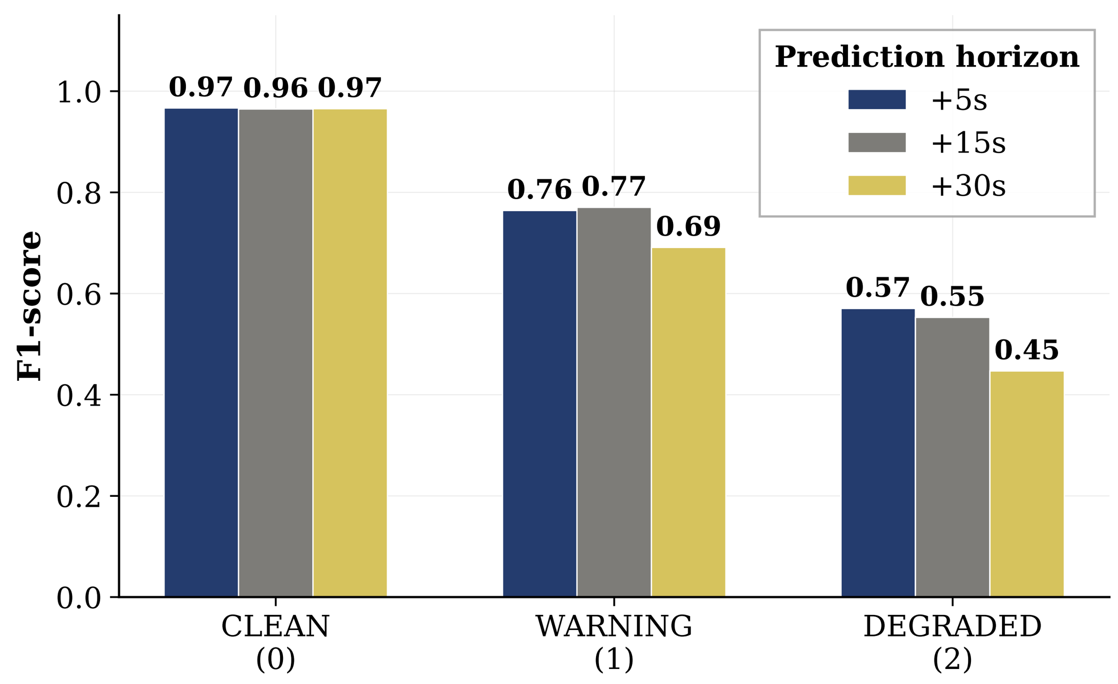

Figure 4: Multi-horizon Macro-F1 and per-class F1. The +5 s horizon is used for real-time EKF integration.

6. Cross-City Generalisation

Trained on Beihang + Hong Kong; tested on **Tokyo — never seen**.

Model	Tokyo Macro-F1	Tokyo DEG F1
SENTINEL (DL)	0.649	0.75
RandomForest	0.618	0.15

5× difference in Tokyo DEGRADED F1. Trees memorise city-specific thresholds; SENTINEL learns transferable degradation representations.

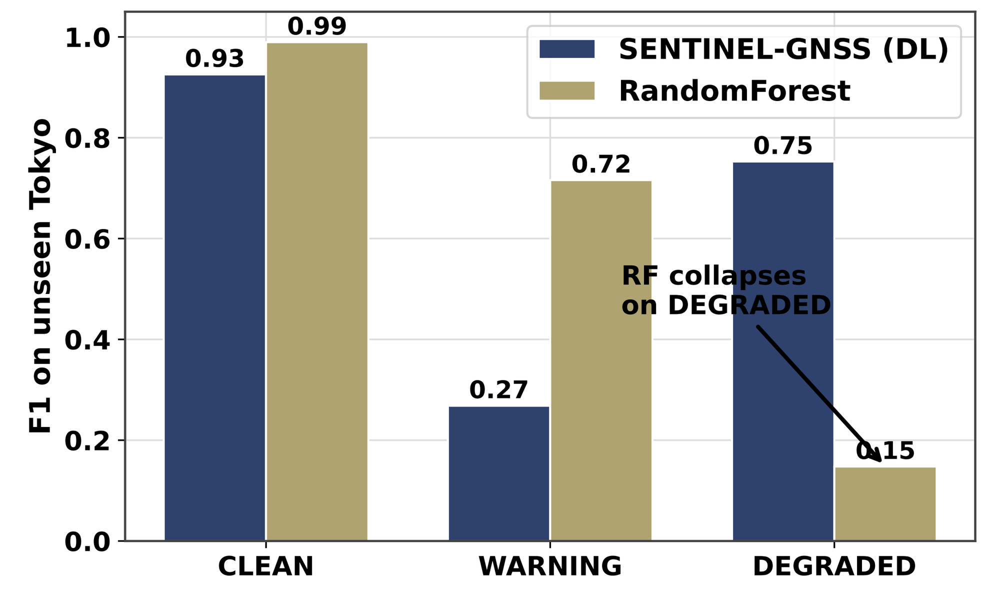

Figure 5: Cross-city DEGRADED F1: DL transfers, RF collapses.

7. Prediction-Informed EKF Fusion

SENTINEL +5 s prediction inflates GPS measurement noise covariance *before* bad measurements arrive:

$$\mathbf{R} = \begin{cases} (40\,\mathrm{m})^2\mathbf{I} & P(\mathrm{DEG}) > 0.45 \\ r_{\mathrm{base}}^2\mathbf{I} & \text{otherwise} \end{cases}$$

Also evaluated: **Huber robust EKF** (IRLS downweighting) and **Bootstrap PF** (Student-t GPS likelihood, N=500 particles).

Method / Scenario	Deg. RMSE	Gain
EKF Fixed-R (Trimble)	24.3 m	**+48.8 %**
Huber EKF (u-blox)	58.6 m	**+25.2 %**
PF Student-t (Odaiba)	35.4 m	**+40.1 %**

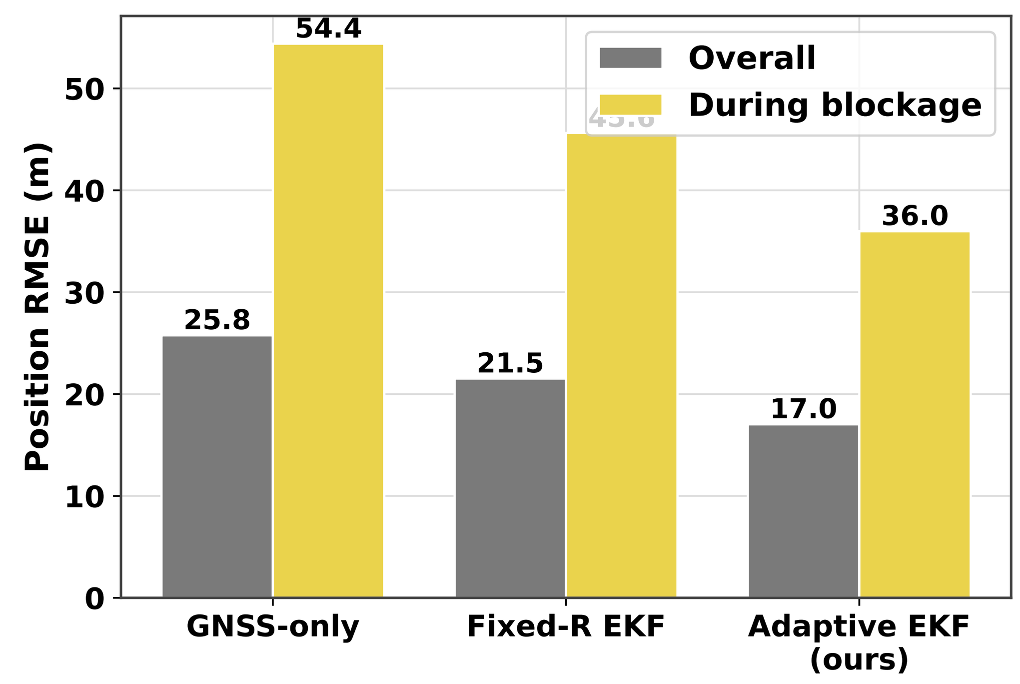

Figure 6: Degraded-segment RMSE: all methods × all scenarios. No single method dominates — environment-aware selection matters.

8. Conclusions

- GNSS degradation is **predictable 5–30 s in advance** with Macro-F1 = 0.821 at the actionable +5 s horizon
- The hybrid Transformer+BiLSTM **transfers across cities**; classical ML does not (5× DEGRADED F1 gap on Tokyo)
- Prediction-informed EKF fusion reduces degraded positioning RMSE by **up to 48.8 %** in real Tokyo urban drives
- Environment-aware method selection: Fixed-R EKF (clean GPS), Huber EKF (heavy-tailed), Student-t PF (random NLOS)

Code: `github.com/Jorshuare/AI-Based-Prediction-for-`

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